

Evolutionary Feedback between Petrologic Transformations, Fluid Transport, and Slab Dynamics in Subduction Models

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Key Points:

- Dynamic subduction models capture feedback between pressure-temperature state, serpentinization-induced weakening, and kinematics.
- Serpentinization affects plate speeds, especially after early subduction.
- Sediment may lubricate the plate interface indirectly through the ingrowth of weak serpentinite.

Abstract

During subduction, the downgoing crust is exposed to high temperatures in the mantle wedge, causing the breakdown of volatile-bearing minerals and releasing fluids into the forearc. These hydrous fluids percolate upwards, reacting with the mantle wedge and forming hydrous ultramafic lithologies, especially serpentinite. To more accurately track the fate and impact of water on the forearc, we develop models that link serpentinite ingrowth to interface weakening. Unlike previous models, which prescribe plate velocities, geometries, or steady-state conditions, our approach allows plates to evolve dynamically without predefined velocities or geometries. Using these dynamic models, we simultaneously track the feedback between temperature and water storage within serpentinites and the resulting impact of weak serpentinite on plate interface strength over time. We primarily consider variations in the strength of serpentinite and hydration of the incoming plate. Fully saturated H_2O in the incoming plate leads to up to a $\sim 3\times$ increase in cumulative serpentinite production compared to anhydrous sediment, resulting in faster convergence rates. Convergence rates are $\sim 40\%$ lower for cases with no incoming sediment, which experience half as much serpentinite ingrowth as cases with fully saturated pelagic sediment. Serpentinite strength modulates convergence rates: serpentinite with lower friction coefficients lubricates the interface more but is also more susceptible to downdragging due to coupling with the downgoing plate. This effect is especially prominent during the free sinking and mature phases of subduction. While our work focuses on serpentinite, our results are broadly consistent with lubrication from metasomatic reactions, such as the formation of chlorite or talc. Previous studies have also suggested that sediments may lubricate the interface. We find that hydrous sediments may not only lubricate the interface directly by being weak, but also indirectly by dehydrating and forming weak serpentinite in the mantle wedge.

Plain Language Summary

We run models of subduction where plate forces and speeds can change freely and where water movement is tracked. Water moving upwards from the plate interface forms weak rocks called serpentinites. We find these weak serpentinites can lubricate the interface between plates and speed up subduction over time. This is also important because previous research proposed that sediments may change subduction speeds. Instead, sediments with water form weak serpentinites or other minerals that could also lubricate the interfaces of subducting plates.

1 Introduction

The mechanical and compositional properties of the plate interface underpin a range of subduction behaviors, including deformation (Shreve and Cloos, 1986; Behr and Burgmann, 2018), rock exhumation (Gerya et al., 2002; Agard et al., 2018), deep mantle hydration (Rüpke et al., 2004; Magni et al., 2014), and volatile budgets in the mantle wedge (Hyndman and Peacock, 2003; Hacker, 2008; van Keken et al., 2011). Volatiles are liberated from the downgoing plate, migrate upwards and, depending on their composition and pressure-temperature (P-T) conditions, may be bound in hydrous minerals, such as serpentinite

(Peacock, 1990). Given the complexity of devolatilization patterns, numerical subduction models are valuable tools to estimate the patterns and magnitudes of dehydration and/or hydration.

Previous studies have successfully modeled dehydration and mantle wedge hydration during subduction. A common method for predicting dehydration patterns has been to couple the thermal fields predicted by kinematically driven subduction models, which have fixed slab dips and convergence rates, with metamorphic phase stability calculations to estimate water flux (Hyndman & Peacock, 2003; Hacker 2008; van Keken et al., 2011; Abers et al., 2017). However, early subduction zone thermal structures that cause dehydration are predicted to cool over time, as the downgoing plate has not yet equilibrated with the mantle geotherm (Platt, 1975; Cloos, 1985; Agard et al., 2018), a trend further supported by the evolving slab thermal state (Peacock, 1990; Molnar and England, 1990; Wang et al., 1995). Evolving thermal conditions, in turn, produce transients in dehydration and associated mantle wedge hydration that cannot be captured in steady-state fixed geometry subduction models. Relatively rapid changes in slabtop temperatures can arise due to dynamic changes in the kinematic properties that make up the “thermal parameter” (e.g., Kirby et al., 1995; van Keken et al., 2010). Convergence rates, for example, can evolve even over few million-year-timescales (Faccenna et al., 2001; Sdrolias and Muller, 2006); this is also borne out within “dynamic” subduction models where plate forces are allowed to evolve freely without external forcing (Clark et al., 2008; Cerpa et al., 2014) and consistently result in highly time-dependent thermal structures and relatively rapid changes in water release (Kincaid and Sacks, 1997; Holt and Condit, 2021; Zhou and Wada, 2021). A one-way postprocessing of the dynamic Holt and Condit (2021) subduction zone thermal models found that water release and storage in mantle wedge serpentinite is greater than that predicted within fixed geometry and constant velocity subduction models (Abers et al., 2017) due to the inclusion of the hotter, earlier stages of subduction (Epstein et al., 2024).

While several previous studies have incorporated time-dependent hydration of the mantle wedge and associated serpentinite ingrowth, they have typically prescribed plate speeds and/or geometries. This assumption neglects the dynamic evolution of subduction zone properties, such as convergence rate and dip, and hence the resulting impact of kinematics on temperatures and slab/wedge dehydration/hydration. Such studies have successfully modeled serpentinitization in contexts such as high/ultra-high pressure rock exhumation (Gerya et al., 2002; Gerya & Stöckhert, 2006), arc magmatism (Gorczyk et al., 2007; Nikolaeva et al., 2008), mantle wedge kinematics (Arcay et al., 2005; Wada et al., 2008; Kerswell et al., 2021), and steady-state water budgets (Wada et al., 2012; Abers et al., 2017). However, dynamic models with freely evolving force balances are needed to capture time-dependent feedback between plate kinematics and interface strength. Such feedback may arise from weak materials that could lubricate the plate interface (Behr et al., 2022), including sediment (Shreeve and Cloos, 1986; Jagoutz et al., 2015; Behr and Becker, 2018) and altered oceanic crust (Kimura and Ludden, 1995), whose quantities generally remain steady over time. By contrast, hydrous minerals, such as serpentine (Moore and Lockner, 2004; Hilairet et al., 2007; Wada and Wang, 2008) or talc (Horn and Deere, 1962; Boneh et al., 2023), may accumulate in the mantle wedge as hydrous fluids are released from downgoing slabs. Their changing quantities may establish

time-dependent feedback with subduction kinematics. Indeed, a full feedback loop can arise because serpentinite accumulation depends on evolving P-T conditions. These P-T conditions are linked to kinematics, such as plate speeds that, in turn, are partially controlled by the strength of weak materials at the plate interface. Therefore, fully assessing the effects of serpentinite on interface strength requires fully coupled, time-dependent dynamic models, because serpentinite quantities and plate interface strength change over time.

Building on this foundation, we present the first attempt to model time-dependent effects of serpentinization in dynamic subduction models. To isolate the impacts of slab dehydration and wedge hydration and associated strength changes, we examine numerical models with varying interface yielding properties and hydration states of the downgoing crust. By simulating serpentinite formation and its feedback on temperature, water release, and plate interface lubrication, these models directly capture and quantify time-dependent subduction zone strength and dynamics.

2 Materials and Methods

We develop 2-D ocean-ocean subduction models using the ASPECT 2.5-pre finite element code (Heister et al., 2017; Kronbichler et al., 2012; ASPECT v2.5.0; Rose et al., 2017; Gassmöller et al., 2018). We solve for the conservation of momentum, mass, and energy, respectively. The incompressible Boussinesq approximation is used for mass conservation, and energy conservation assumes adiabatic heating. There are no contributions of radioactive or shear heating:

$$\begin{aligned} 1. \quad & \nabla \cdot 2\eta \dot{\mathbf{e}}(\mathbf{v}) + \nabla p = \rho \mathbf{g} \\ 2. \quad & \nabla \cdot \mathbf{v} = 0 \\ 3. \quad & \rho C_p \left(\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T \right) - k \nabla^2 T = H \end{aligned}$$

Individual lithologies are represented by compositional fields that are advected by the flow field. Advection of these compositional fields is treated by adding an additional equation of advection (Equation 4) for each field c to the preceding equations (Equation 1-3). Stabilization of these ancillary advection equations is achieved by introducing an artificial diffusivity v_h in the equation (Glerum et al., 2018):

$$4. \quad \frac{\partial c_i}{\partial t} + \mathbf{v} \cdot \nabla c_i - \nabla \cdot (v_h(c_i)) \nabla c_i = 0$$

For our geometry, we consider a 90-million-year-old oceanic plate subducting below younger oceanic crust. Spreading centers are located 500 km from either boundary of the domain at the margins of both the subducting plate and overriding plate. An initial curved section of the subducting plate is included that extends to a depth of 50 km and to 145 km onboard of the trench. The model dimensions are 5800x1450 km. All other associated parameters are listed in Table 1 and Table S1.

The models are dynamic, with plate forces and geometries allowed to evolve freely and, in terms of thermal structure and (non-interface) mechanical properties are set up the same as Holt and Condit (2021). Our models contain a free surface at the upper boundary and are

allowed to isostatically equilibrate for 100 timesteps before being run for 50 million years (Figure 1). Surface diffusion uses a diffusion coefficient k of 10^{-7} with free surface stabilization. Except for the free surface upper boundary, free-slip boundaries are enforced elsewhere. A constant background mantle temperature of 1421°C is used (GDH1 plate cooling model: Stein & Stein, 1992). Constant temperatures are enforced at the bottom (1421°C) and top (0°C) of the domain. Insulated (no heat flux) boundary conditions are enforced on the left and right sides of the domain.

ASPECT's adaptive mesh refinement (AMR) functionality is used to locally increase model resolution in regions with high strain rates, crustal compositions, high temperatures, or with cutoffs based on quantities of crustal compositions. AMR levels in ASPECT are formulated based on a subdivision of the domain in some dimension to powers of two. A minimum refinement of 5 ($2^5 = 32$ subdivisions) is used for a minimum resolution of ~ 230 km and an additional 5 adaptive refinements ($2^{10} = 1024$) are used for a maximum resolution of ~ 1.3 km. In the reference model the initial number of degrees of freedom is $\sim 2,900,000$, and the final number of degrees of freedom at the end of the run is $\sim 9,400,000$. A minimum refinement function is used to resolve the 660 km discontinuity with a grid size of ~ 2.6 km enforced within 2 km of the interface. All parameters associated with AMR are listed in supplemental tables S2-S3.

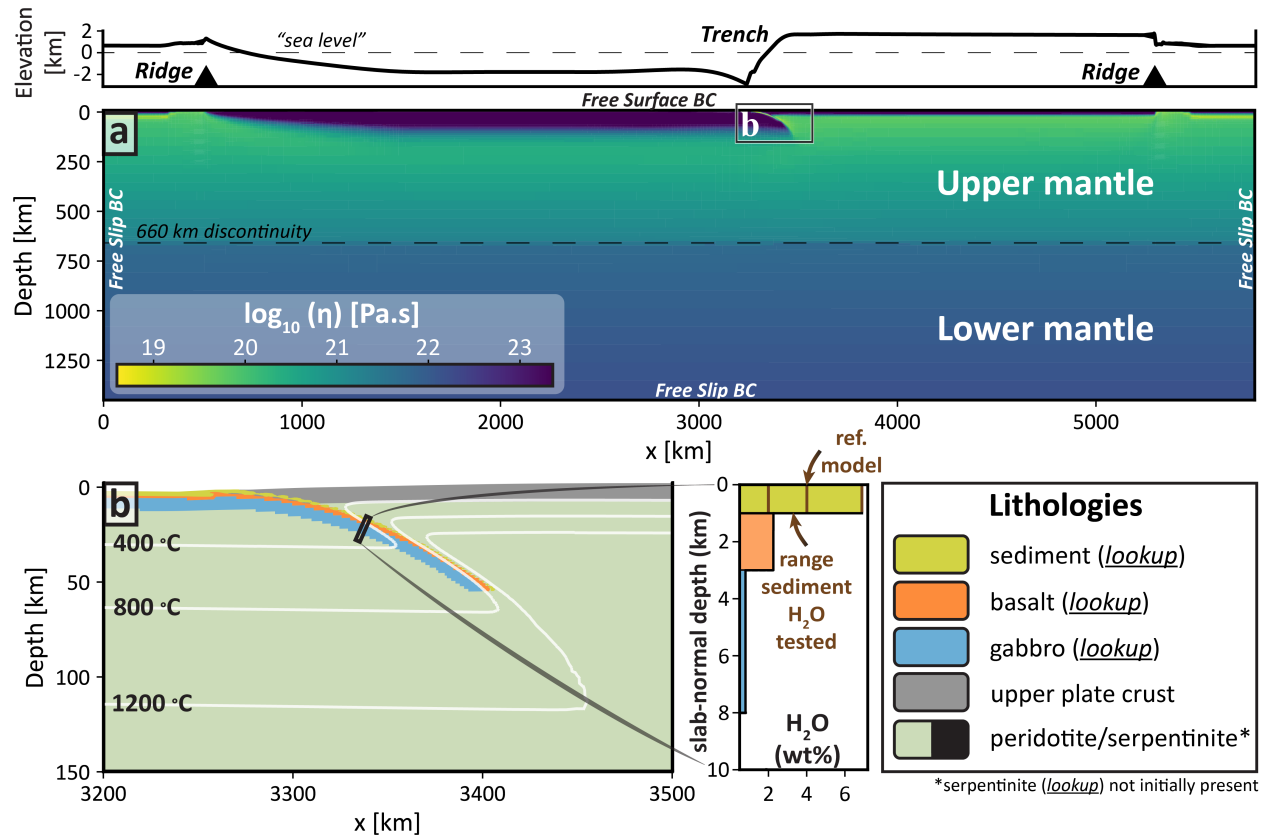


Figure 1. Model set-up showing the initial viscosity field and initial, isostatic topography (a) with a zoom-in of the subduction interface region (b). The lithologies/compositions of the downgoing crust are also shown, with

thicknesses and the range of explored sediment H₂O concentrations (see text Methods section 2.2). As illustrated, some compositions use lookups to assign rheology and density (see Table 1).

2.1 Phase stability

Our modeled crust is constructed as a simplified version of the one described in Hacker (2008) and Jarrard (2003; Figure 1b). The crust is composed of, in order of increasing depth away from the slabtop, 1 km of pelagic sediment, 2 km of metabasalt, and 5 km of metagabbro. Assuming a 90 Myr old oceanic crust, the metabasalt H₂O content is calculated as an average of the H₂O content of 300 m of upper volcanic extrusives (6.04 wt. %), 300 m of lower extrusives (4.1 wt. %) and 1.6 km of sheeted dykes (1.76 wt. %) for an H₂O content of 2.3 wt. %. The gabbro has 0.79 wt. % H₂O, and the pelagic sediment 6.89 wt. % H₂O when fully saturated, respectively. We assume a composition based on van Keken et al. (2011) for 1 km of pelagic sediment. No initial pore fluid is included for any of the crust (Table 1). Hydration of abyssal serpentinite is not considered.

Metamorphic phase diagrams are used to track the stability of minerals in individual compositions, which evolves as a function of pressure-temperature (P-T) conditions. Density, viscosity, and fluid release for all crustal compositions depend on the results of phase stability calculations unless otherwise stated. Fluids, assumed here to be exclusively H₂O, are tracked, and metamorphic phase diagrams are used to calculate the amount of H₂O stable in hydrated compositions similarly to previous studies (van Keken et al., 2011; Wada et al., 2012; Abers et al., 2017; Epstein et al., 2024). Specifically, the phase stability and postprocessing is calculated using Perple_X version 7.0.1 (Gabe?) (Connolly, 2005). All thermodynamic calculations were carried out in the system Na₂O-CaO-K₂O-FeO-MgO-Al₂O₃-SiO₂-H₂O. We use the thermodynamic database of Holland and Powell (2011) and Pitzer and Sterner (1994) for fluid equation of state. Solid solution models for all calculations are in the supplemental material (Table S4). A maximum decoupling depth (MDD) of 200 km is chosen. At depths greater than the MDD, the downgoing crust has properties identical to the background mantle (cf. Holt and Condit, 2021).

2.2 H₂O Transport

Although the mantle wedge is not initially hydrated, hydrated crust from the downgoing plate encounters relatively hotter conditions in the mantle wedge, causing the crust, including the pelagic sediment, to dehydrate and release free H₂O, or water. We use “water” throughout the text interchangeably with H₂O even though at P-T conditions here this fluid may be supercritical. Figure 2 shows a flowchart depicting the treatment of water release, transport, and storage in the model. After the 0th timestep, bound H₂O is assigned in all compositions where water is expected to be present (i.e., following Figure 1b). At each subsequent timestep a new stable bound H₂O is calculated using lookup tables from PerpleX. The new bound H₂O compared to the current H₂O. new bound H₂O is then the current bound H₂O, the remaining is transformed into a “free H₂O” phase. Treating compositions as a continuum and not as particles means that they can be less than unity and that individual compositions in the crustal column can diffuse into each other (Kronbichler et al., 2012). We therefore scale each H₂O amount for

the magnitude of each composition (from 0-100%) within a given cell to prevent over or under-prediction of the H₂O content.

The free H₂O is then, for simplicity, advected vertically upward at a constant rate of 10 cm/yr (e.g., Gerya et al., 2002). In practice, this is implemented by modifying Douglas et al.'s (2023) ASPECT “plug-in” that adds two-phase (Darcy) flow to ASPECT. Here, instead of solving for the fluid velocity, we set it at a constant value in the plug-in.

Free H₂O is removed from the model domain unless it interacts with the mantle wedge. It may interact with the mantle wedge in three ways: a) if the temperature is above the calculated wet solidus (Katz et al., 2003) then all the free fluid is assumed to be absorbed into the melts, which can readily accept 10s of wt. % H₂O (Gabe?). The melt migration and the effect of melt on viscosity are not modeled: a melt region is calculated purely to be consistent with H₂O being incorporated into melts, remove the associated free H₂O, and hence to prevent extensive serpentinization of the base of the overriding plate; b) If the temperature is below that of the solidus yet above that of serpentinization stability, free H₂O continues to be advected upwards but weakens the surrounding mantle by a factor of 100. This mimics the weaker rheology of a hydrated peridotite, which has a lower viscosity than dry peridotite (Hirth and Kohlstedt, 2003). c) If the free H₂O is in the mantle and within the serpentine stability field, it is bound within a “serpentine compositional field.”

Serpentine can, however, be dragged downwards to temperatures greater than those of serpentine stability. While the resulting deserpentinization is not tracked directly, if any such material in the serpentine field exits the region of serpentine stability, then the background olivine flow law is re-imposed in the lookup. We will use the terms *cumulative serpentine* to denote the “serpentine” compositional field, which includes both currently stable serpentine and mantle material that has deserpentinized, and the term *instantaneous serpentine* which refers to any serpentine that is currently stable as serpentine.

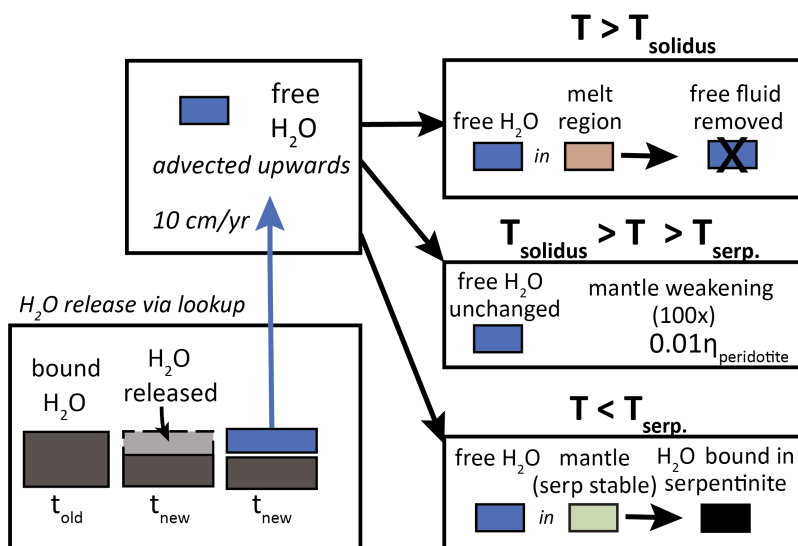


Figure 2. Flowchart illustrating our implementation water release, transport, and storage. Bound H₂O is calculated with lookups from PerpleX. At each time step new and old H₂O contents are calculated, and the remaining is

transformed into a “free H₂O” phase, which is advected upward at 10 cm/yr. If free H₂O interacts with the mantle wedge it: a) is absorbed into melts if the temperature is above the calculated wet solidus H₂O (Katz et al., 2003), b) continues to be advected upwards, but causes a weakening of the surrounding mantle by 100x, if the temperature is between the temperature of the solidus and serpentinization, c) is bound within a “serpentinite field” if serpentinite is stable based a separate lookup.

2.3 Rheology

We adopt a composite flow law, with diffusion creep, dislocation creep, and pseudo-plastic yielding for the background lithosphere and mantle (i.e., outside of the plate interface materials and any serpentinite). Diffusion and dislocation creep mechanisms follow the idealized flow law:

$$5. \quad \eta_{creep} = A^{\frac{-1}{n}} \dot{\epsilon}^{\frac{1-n}{n}} f_{H_2O}^r \exp\left(\frac{E+PV}{nRT}\right)$$

where A is the creep prefactor, n is the stress exponent (1 for olivine diffusion creep; $\approx$ 3.5 for olivine dislocation creep), $\dot{\epsilon}$ is the second invariant of the strain rate tensor, E is the activation energy, V the activation volume, R the ideal gas constant, and T the temperature (Table 1).

Composition	Flow law ^a	A [Pa·n·s ⁻¹]	n	E _a [kJ/mol]	r	V [cm ³ mol ⁻¹]	τ_0 (LTP) ^b [GPa]	p (LTP) ^b	q (LTP) ^b	melt	CO	meff MPa	Reference(s)	Initial H ₂ O (wt%)	Reference density (kg/m ³)
sediment	wet quartz	6.31x10-36	4	135	1	0	—	—	—	—	—	0.02; 0.04	Hirth et al. (2001)	6.89	lookup
metabasalt	wet quartz	6.31x10-36	4	135	1	0	—	—	—	—	—	0.02; 0.04		2.32	lookup
	amphibolite	6.94x10-27	3.7	244	0	0	—	—	—	27	—				
	eclogite	1.99x10-18	3.5	403	0	27.2	—	—	—	27	—				
metagabbro	wet quartz	6.31x10-36	4	135	0	0	—	—	—	—	—	0.02; 0.04		0.79	lookup
	amphibolite	6.94x10-27	3.7	244	1	0	—	—	—	27	—				
	eclogite	1.99x10-18	3.5	403	0	27.2	—	—	—	27	—				
serpentinite	serpentinite ^{bc}	4.37x10-22	2	86.3	—	—	2.42	1	1.18	—	—	0.01- 0.06	Burdette & Hirth (2022); Fildes & Billen (2025)	—	lookup
initial weak zone	same as metabasalt	—	—	—	—	—	—	—	—	—	—	0.005		0	3350
mantle	dry olivine (disl. creep)	9.153x10-20	3.5	540	0	12	—	—	—	—	—	0.36		0	3300
	dry olivine (diff. creep)	9.6x10-12	1	300	0	4	—	—	—	—	—			0	3300
overriding plate oceanic crust	same as mantle	—	—	—	—	—	—	—	—	—	—	0.36		0	2800
yielding region	same as mantle	—	—	—	—	—	—	—	—	—	—	0		0	3300

lower mantle	20x olivine diff. creep	4.8x10 ⁻¹³	1	300	0		—	—	—	—	0.36		0	3300
^a Equation 5														
^b Fildes & Billen, 2025														
^c Equation 11														

Table 1. Model parameters associated with individual compositions.

We also incorporate pseudo-plastic yielding in the form of a “plastic” viscosity, $\eta_{plastic}$. The process of forming brittle fractures at the scale seen in nature, on the sub-meter scale or outcrop scale (e.g., Rowe et al., 2013), is computationally infeasible in the current models. Furthermore, multiple strain mechanisms, brittle, semi-brittle, and viscous may be operating concurrently. All such deformation processes tend to produce highly localized regions of shear and are appropriate for the bending lithosphere. We therefore subsume all such mechanisms into this plastic component using a 2-D Drucker-Prager criterion to prescribe a yield stress, σ_y :

$$6. \quad \sigma_y = C \cos(\varphi) + \sin(\varphi)P$$

Where C the cohesion, P the pressure, and a friction angle φ . Furthermore, an array of studies has already shown that pore-fluid pressure is likely high during shear (Audet & Burgmann, 2014; Peacock et al., 2011; Condit and French, 2020). We use an “effective friction” coefficient that is meant to include the effects of pore-fluid overpressure, which serves to weaken the interface. We therefore recast equation 6 as:

$$7. \quad \sigma_y = \mu_{eff}P + C_0$$

$$8. \quad \mu_{eff} = (1 - \lambda)\mu$$

C_0 is a modified cohesion coefficient that includes the friction angle term, and λ is a pore fluid pressure term. The yielding viscosity is treated by setting an effective plastic viscosity as a function of the yield stress and effective deviatoric strain rate $\dot{\epsilon}_e$:

$$9. \quad \eta_{eff}^{pl} = \frac{\sigma_y}{2\dot{\epsilon}_e}$$

Viscosities associated with the different rheological mechanisms are then harmonically averaged to generate the model (or “composite”) viscosity:

$$10. \quad \frac{1}{\eta_{composite}} = \frac{1}{\eta_{diff}} + \frac{1}{\eta_{disl}} + \frac{1}{\eta_{plastic}}$$

The subscripts “diff,” “disl,” and “plas” refer to diffusion creep, dislocation creep, and plastic viscosity components. If a model grid cell contains two compositions (e.g., basalt and sediment), the respective composite viscosities are averaged using a geometric mean.

2.3.1 Mantle Rheology

The viscous component of the background mantle deforms via a combination of diffusion and dislocation creep, with activation volumes and energies corresponding to those of dry olivine (Karato & Wu, 1993; Table 1). Diffusion and dislocation prefactors are set to produce a composite viscosity of 2.5×10^{20} Pa s (i.e., $\eta_{\text{diff}} = \eta_{\text{disl}} = 5 \times 10^{20}$ Pa s) with a strain rate of 10^{-14} s⁻¹ at a background mantle temperature and lithostatic pressures of 330 km depth (e.g., Billen and Hirth, 2005).

These values result in sub-lithospheric deformation that is dominated by dislocation creep down to depths of ~ 250 km, as is broadly consistent with constraints from seismic anisotropy (Becker, 2006; Podolefsky et al., 2004). At depths greater than 660 km, the lower mantle deforms only via diffusion creep, with the lower mantle diffusion creep prefactor set to produce a viscosity 20 times that of upper mantle diffusion creep. Our experiments are presented as “effective” friction coefficients to denote friction values that consider both friction μ , as traditionally measured in rock deformation experiments and a pore fluid pressure factor $(1-\lambda)$ that modulates yielding ($\mu_{\text{eff}} = \mu(1-\lambda)$). Therefore, the friction angle associated with plastic yielding is set to produce a μ_{eff} of 0.36 (i.e., considering hydrostatic conditions with $\lambda = 0.6$ and $\mu = 0.6$) and a cohesion of 20 MPa. Additionally, to keep the spreading centers active we include a weak box with a cohesion of 1 MPa. All associated rheological parameters for compositions are listed in Table 1.

2.3.2 Crustal Rheology

For oceanic compositions, we use lookup tables (metabasalt, metagabbro) to impose the flow law associated with the weakest mineral predicted to be stable at those pressure-temperature conditions (Figure S1). In metamafic lithologies dislocation creep flow laws are used for quartz (Hirth et al., 2001), amphibolite (Hacker & Christie, 1990), and eclogite (Zhang & Green, 2007). We adopt a dislocation creep quartz flow law for the pelagic sediment (Hirth et al., 2001).

For plastic yielding we set the friction angle so that a μ_{eff} of 0.36 is used for metagabbro and upper plate crust. Interface rheologies tend to have high values of the pore fluid pressure coefficient λ , consistent with overpressures that would significantly weaken the interface. For example, the orientations of crack-seal veins at conditions 300-350 °C in a seismogenic zone is consistent with high overpressures and λ values >0.95 (Condit and French, 2022). Therefore, for metabasalt and pelagic sediments yielding is explored in the range 0.02-0.04 (i.e., considering $\lambda \sim 0.95$ and $\mu = 0.6$). An initial weak seed ($\mu_{\text{eff}} = 0.005$) in the curved section of the plate is used to initiate subduction. A cohesion value of 1 MPa is used for pelagic sediment and metabasalt and 20 MPa for metagabbro (Table 1).

2.3.3 Serpentine Rheology

Serpentine deformation deforms via a Peierls creep (low-temperature plasticity; Burdette & Hirth, 2022). Peierls creep is implemented as a dislocation creep approximation of the Peierls creep flow-law (see Fildes & Billen, 2025):

$$\dot{\epsilon} = A \left(\frac{\sigma}{\mu} \right)^2 \exp \left(\frac{-\Delta F}{RT} \left(1 - \left(\frac{\sigma}{\tau_0} \right)^p \right)^q \right)$$

At each timestep, the stability field of serpentinite is checked based on a lookup table, and if the composition is within serpentinite stability, a serpentinite flow law is used. Outside of serpentinite stability we adopt the rheological properties of the background mantle. Water that would be re-released by deserpentinizing serpentinite is not tracked.

Serpentinite yielding parameters generally have a range of values, depending on serpentine species, temperature, velocity, and pressure. For example, friction coefficients (μ) of lizardite range from 0.18 (Carpenter et al., 2009; Tesei et al., 2018) to ~ 0.55 (Dengo and Logan, 1981; Moore and Lockner, 1997), with many experiments falling in the 0.25-0.4 range (Moore and Lockner, 2004; Behnsen and Faulkner, 2012; Scuderi and Carpenter, 2022). The high temperature behavior of serpentine species may also be weaker. The high temperature polymorph of serpentine, antigorite, generally has a higher coefficient of friction of 0.5-0.85 (Raleigh & Patterson; Reinen et al., 1994), which may be decreased if dissolution-precipitation creep enhanced failure (Rutter 1976; Behr and Platt, 2013; Tulley et al., 2024). Therefore, the friction angle associated with plastic yielding is set so that a μ_{eff} is explored in the range 0.01-0.06 with a cohesion of 1 MPa (i.e., considering $\lambda \sim 0.90$ -0.95 and $\mu = 0.3$ -0.6; Table 1).

3 Results

We first present our reference subduction model that includes serpentinite ingrowth over time. Next, we analyze similar models with variable H_2O inputs into the subduction zone, including with and without a sediment layer. Finally, we vary serpentinite strengths to test the sensitivity of kinematics and pressure-temperature (P-T) conditions to accumulating weak serpentinite.

3.1 Reference model

Because the reference model is dynamic, as are all the models, the boundary conditions are not prescribed and hence subduction geometries and velocities are allowed to evolve freely through time thereby impacting, and being impacted by, serpentinite ingrowth for 50 Myr (Figure 3). The co-evolving convergence rates, serpentinite content (Figure 4a) and slab pressure-temperature (P-T) profiles (Figure 4b, c) are examined. In Figure 4, we overlay these P-T profiles on the pseudosections used for pelagic sediment and metabasalt, the two most hydrated compositions.

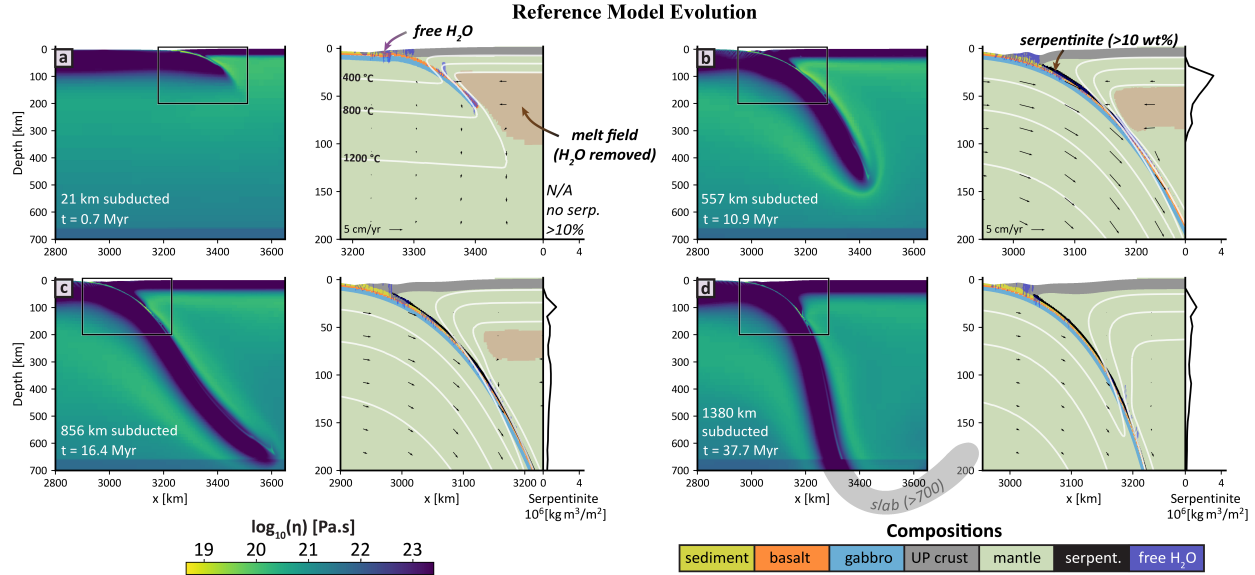


Figure 3. Snapshots of reference model evolution over time. (a, b) Model stages showing the initiation and free-sinking phases, respectively, with the downgoing slab releasing free H_2O that hydrates the mantle wedge via serpentinization. (c, d) Evolution before and after the 660 km transition zone. (d) Grey shows extent of slab outside zoom-in region. To limit excess hydration of the overriding plate, a hydrous "melt" region is calculated (Katz and Spiegelman, 2003), where free H_2O is assumed to be incorporated into melt and removed from the system (see Methods, Section 2.2, for details).

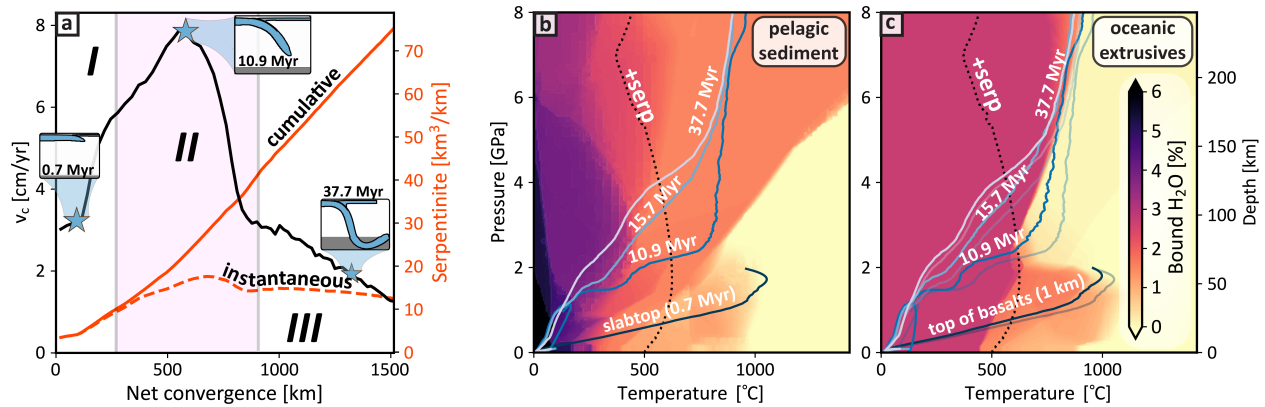


Figure 4. Kinematic and thermal evolution of the reference model. (a) Ingrowth of cumulative serpentinite produced over the course of the model, and instantaneous serpentinite content including serpentinite stable at a given point in time, i.e. without deserpentinized/downdragged material. I refers to the initiation stage, II to the free sinking phase, III to the mature phase as classified by previous studies (e.g., Holt and Condit, 2021). (b-c) Evolution of slabtop and near slabtop thermal state. Background shows bound water used in lookups for pelagic sediments and oceanic extrusives, respectively. Lightly shaded P-T paths (c) coincide with the slabtop P-T, and darker shading coincides with P-T conditions at the top of the metabasalt section.

During the *initiation stage*, until ~ 8 Myr, (I) the slab is near its initial position and largely sinks near-vertically (Figure 3a). During this time subduction is sluggish but accelerating from ~ 3 to 6 cm/yr (Figure 4b). Convergence is dominantly accommodated via rollback, with overriding plate motion initially $\sim 5\times$ faster than that of the subduction plate. This initiation stage is the hottest phase, with the slabtop significantly warmer than during the latter stages (Figure 4c). As a result, H_2O is released from the downgoing crust into the mantle wedge up to a shallow depth

of ~ 60 km. Released H₂O migrates upwards and can be bound in serpentinite in the mantle wedge, provided serpentinite is stable (Figure 4c). During this initiation phase, serpentinite released over the total model run—*cumulative* serpentinite—is approximately equal to the *instantaneous* serpentinite—the amount stable in the mantle wedge at any given point in time. That is, there is minimal deserpentinization during the initiation stage.

During the following *free sinking phase*, the tip of the downgoing plate approaches the 660 km transition at a steep dip (Figure 3b) before flattening out to a sub-horizontal angle (Figure 3c). At this point, subduction velocities reach their maximum values (II, Figure 4a), and higher advection rates of the incoming slab cause temperatures to cool, as seen in previous time-dependent models (Wang et al., 1995; Kincaid & Sacks, 1997; Holt & Condit, 2021). As a result of cooler temperatures, water release can occur at greater depths than previous phases, broadly coincident with the depths of dehydration from the loss of stilpnomelane and other, hydrous phases (Figure 3b) in pelagic metasediments. Water freed by dehydration causes serpentinite to accumulate in the mantle wedge. However, serpentinite also begins to be dragged downwards, outside of serpentine stability, stabilizing the amount of serpentinite in the mantle wedge (Figure 4b). Finally, feedback loops become activated: and any regions with above 10 wt. % serpentinite experiences reduced viscosity.

During the *mature phase* (III), the slab sinks into the lower mantle, past the 660 km transition zone (Figure 3c-d). After ~ 17.7 million years, convergence rates decrease, and viscosity increases. In this phase, the convergence rate approaches zero as subduction stalls, caused by the slab penetrating the highly viscous transition zone. The thermal field is too cold for metabasalts to dehydrate within the serpentinite field, and any additional dehydration must come from metasediments (Figure 4c). At this point, melting in the wedge is suppressed and serpentinite from the wedge continues to be dragged down by the slab. Downdragging causes the amount of serpentinite in the wedge to decrease slightly, but instantaneous serpentinite content remains roughly unchanged past ~ 17.7 million years (Figure 4a).

3.2 Variable H₂O input

To capture the effect of bound H₂O content on slab dehydration and forearc serpentinization, we vary water content in the downgoing 1 km-thick sediment, comparing a “fully saturated” case with the maximum amount of mineralogically bound H₂O (6.89 wt. %), an intermediate, moderately hydrated case (4 wt. % H₂O), and a completely anhydrous case with 0% H₂O (Figure 5a-c). Additionally, we include a limiting case identical to Figure 5a-c but with no sediment layer above the metabasalts (Figure 5d). In both the anhydrous and no-sediment cases, H₂O release is exclusively from metabasalts and metagabbros.

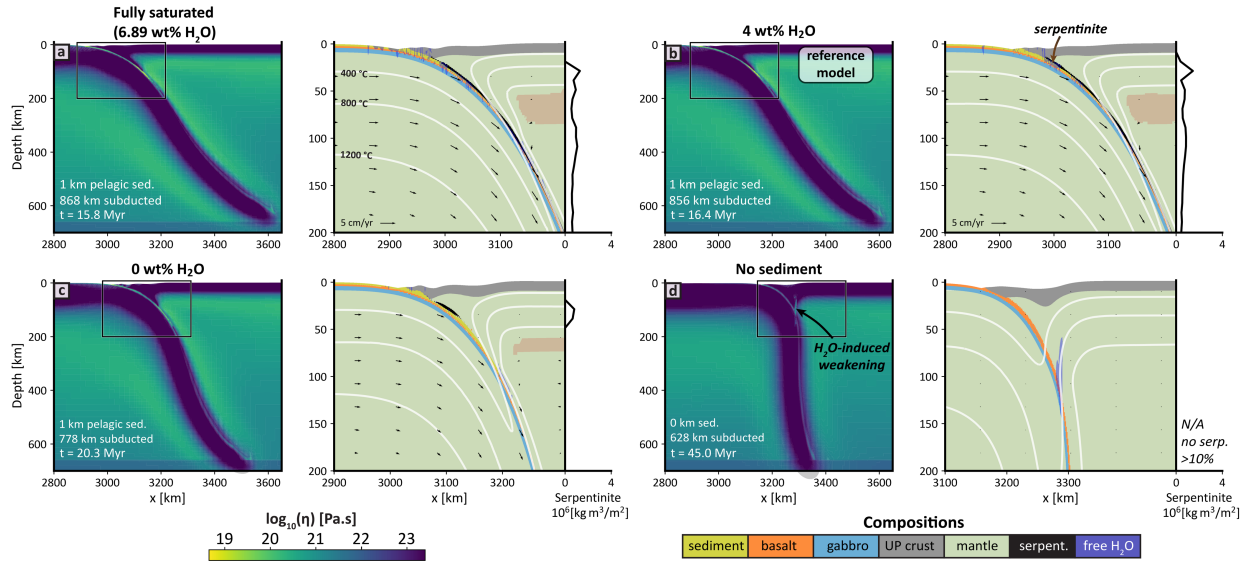


Figure 5. Snapshots of models while varying the hydration state of the incoming sediment. To standardize the amount of bound water input into the subduction zone, all time slices are shown for the same distance subducted, approximately enough to reach the 660 km transition zone. (a-c) Models showing variation in sediment hydration state from fully hydrated to anhydrous (6.89% to 0%). (d) No sediment but with other lithologies still hydrated initially.

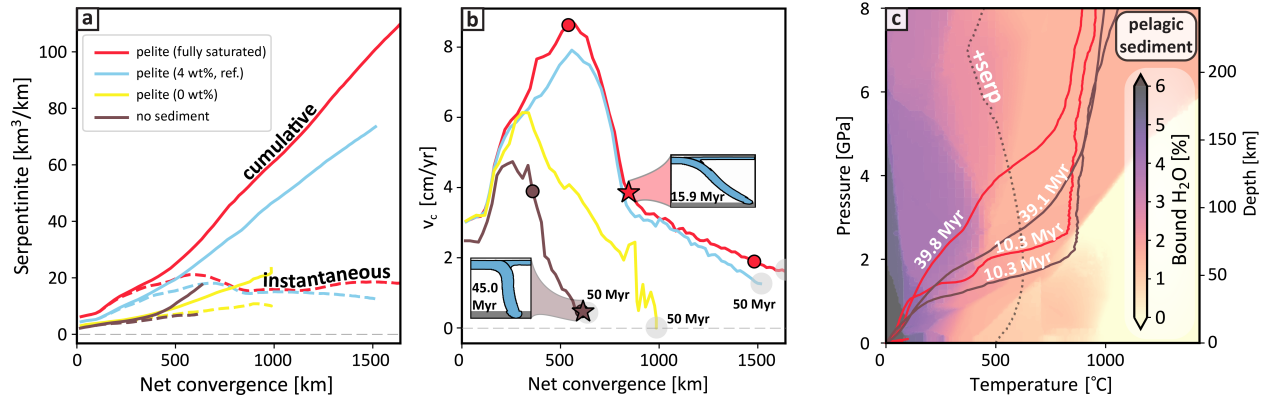


Figure 6. Kinematic and thermal evolution of models with varying water content in sediment or without sediment. (a) Ingrowth of serpentinite over the course of the model run either considering the cumulative serpentinite amount that was created over the duration of the entire model run or the instantaneous amount that indicates the amount of serpentinite that is stable at any given point in time. (b) Plate convergence rates for different models. Filled circles denote times when slabtop P-T conditions are plotted in c. (c) Thermal state at two snapshots for the fully saturated and no sediment cases overlain on the pseudosection for pelagic sediment.

To ensure consistency in comparing serpentinite ingrowth across simulations, we compare models at timesteps with equivalent total amounts of subducted plate length (i.e., *net convergence*) and therefore equivalent amounts of net bound H₂O input from the downgoing slab (i.e., beneath the variably hydrated sediment layer). Specifically, we compare models for a net convergence corresponding to approximately that required for the downgoing plate to reach the higher viscosity lower mantle (i.e., net convergence ~ 860 km). For completeness, we also present a comparison at an equivalent model time (Figure S2).

420 As the incoming plates with different amounts of incoming bound water are exposed to
421 high temperatures in the mantle wedge, they release water that can migrate into the mantle
422 wedge to produce serpentinite. The highest amount of serpentinite is produced in the fully
423 saturated case, both for total *cumulative* serpentinite, which includes deserpentinized material,
424 and *instantaneous* serpentinite that excludes material dragged to P-T conditions beyond
425 serpentinite stability. In the fully saturated case, there is therefore the most serpentinite
426 available to lubricate the interface (Figure 5a). However, the moderately hydrated case and
427 anhydrous (0 wt.%) case both have lower instantaneous and cumulative serpentinite amounts
428 than the fully saturated case, and therefore they experience more coupling (Figure 5a-c).

429 In the case without a sediment layer, sediment is not insulating the interface basalt
430 layer and so the hydrated oceanic extrusives are directly exposed to the hotter temperatures in
431 the mantle wedge (Figure S3). As a result, the case with no sediment is hotter, dehydrates
432 more, and therefore produces more cumulative serpentinite than the anhydrous case. The case
433 without a sediment layer, however, is more coupled and therefore drags down more
434 serpentinite, which acts to reduce stable serpentinite in the wedge to levels comparable to the
435 case with a layer of sediment. Higher coupling arises because metabasalts of the oceanic crust
436 have a strong, eclogite flow law above ~ 1.5 GPa, whereas sediments are governed by a weaker,
437 quartz flow law. Higher coupling in the case with no sediment also produces the highest dip.

438 Convergence rates and thermal state depend on model stage, hydration state, and/or
439 the presence of sediment. As seen in previous models (e.g., Čížková & Bina, 2013; Behr and
440 Becker, 2018; Behr et al., 2022), less coupling along the interface results in faster convergence
441 rates. During the initiation stage, all convergence rates evolve similarly and are accelerating,
442 although the case without sediment is ~ 0.5 cm/yr slower at the model start. After the initiation
443 stage, the case with the most incoming water has the highest degree of serpentinite production
444 and experiences the fastest convergence rates (~ 8.5 cm/yr), whereas the anhydrous case
445 reaches maximum convergence rates of (~ 6 cm/yr). Cases with no water in the pelagic
446 sediment only reach a maximum convergence rate of ~ 5 cm/yr and become fully locked by the
447 end of the model run. This locking is a result of high coupling and high viscosity along interface.
448 Viscosities around the no-sediment interface are generally higher and convergence rates slower
449 compared to the models with sediment (Figure 5). The variation in slab dip is linked to
450 differences in the amount of water release and serpentinite formed (Figure 6). As the slab
451 without sediment reaches the highly viscous 660 km transition zone, the convergence rates
452 slow to <1 cm/yr, locking up by the end of the run due to the high strength of the interface and
453 dramatically increasing the dip. Therefore, the ingrowth of weak serpentinite produces slabs
454 with shallower dips and higher convergence rates, whereas the case with no water has more
455 coupling, higher dips, and higher convergence rates.

456 Thermal state is, in turn, linked back to kinematics. The models with faster convergence
457 rates have a cooler slabtop temperature due to more rapid downward advection of cold
458 lithospheric plate. The case without sediment, with slow convergence, therefore, has a warmer
459 slabtop (Figure 6c).

3.3 Yield Strength

Finally, we test the effect of serpentinite yield strength on model evolution. Snapshots of the slab morphology for models with serpentinite strength are again plotted for equivalent net plate convergence (Figure 7). We vary yield strength values by adjusting the coefficient of friction μ_{eff} above and below the reference value of 0.02 (Equation 7) ranging from 0.01 to 0.04. Varying serpentinite strength, within these bounds, causes a relatively subtle change in slab behavior: a higher coefficient of friction leads to a higher dip of the slab (Figure 7). The *cumulative* amount of serpentinite created over the whole model duration is approximately the same for any amount of convergence regardless of serpentinite strength (Figure 8a).

During the initiation stage (<400 km) all models evolve similarly, with little influence of serpentinite strength on downdragging (Figure 8a-b). After the initiation stage, however, convergence rate and serpentinite retention are very sensitive to friction: weak forearc serpentinites with low friction coefficients are dragged down more easily than those with high friction coefficients, despite roughly similar cumulative serpentinite ingrowth. That is, the serpentinite stable at any given time—the *instantaneous* serpentinite—is affected by downdragging. During the free sinking phase there is 50% more instantaneous serpentinite in the high friction case compared to a case with lower friction (Figure 8a). An even higher friction coefficient of 0.06 was separately tested that resulted in the most serpentinite retention and least downdragging (Figure S4).

Changes in instantaneous serpentinite are linked to changes in thermal state and kinematics. The lowest friction case reaches maximum convergence rates of 10 cm/yr, whereas the highest friction case has maximum convergence rates 40% lower than the lowest friction models (Figure 8b), even though there is more serpentinite stable in the high friction case. The thermal state is hotter in the higher friction case not because of thermal dissipation, which is not considered here, but because there are higher velocities and therefore there is less advective cooling of the mantle wedge by the cold slab. Even though the slab is colder in the faster, weaker serpentinite cases, the downgoing slab passes through the same dehydration reactions and therefore, in the long-term, the cumulative amount of serpentinite is comparable to the stronger serpentinite cases. The slabtop in both the low and high friction cases becomes cooler as the subduction zone matures (Figure 8c).

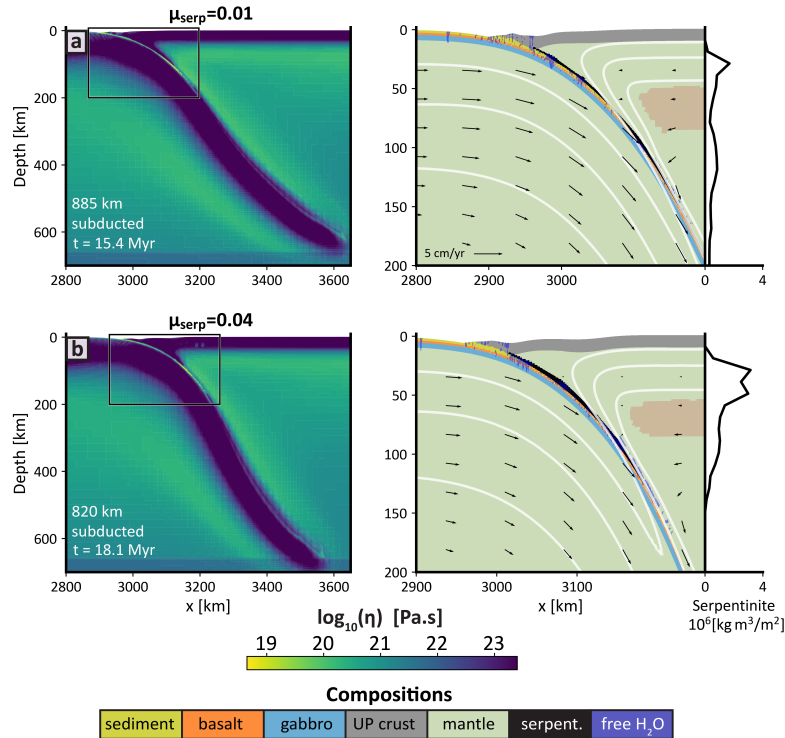


Figure 7. Snapshots of the subducting plate as a function of friction parameters in serpentinite with all other parameters identical to the reference model, shown in Figure 3. All cases are plotted at the same time step, approximately the time for the subducting plate to reach the transition zone.

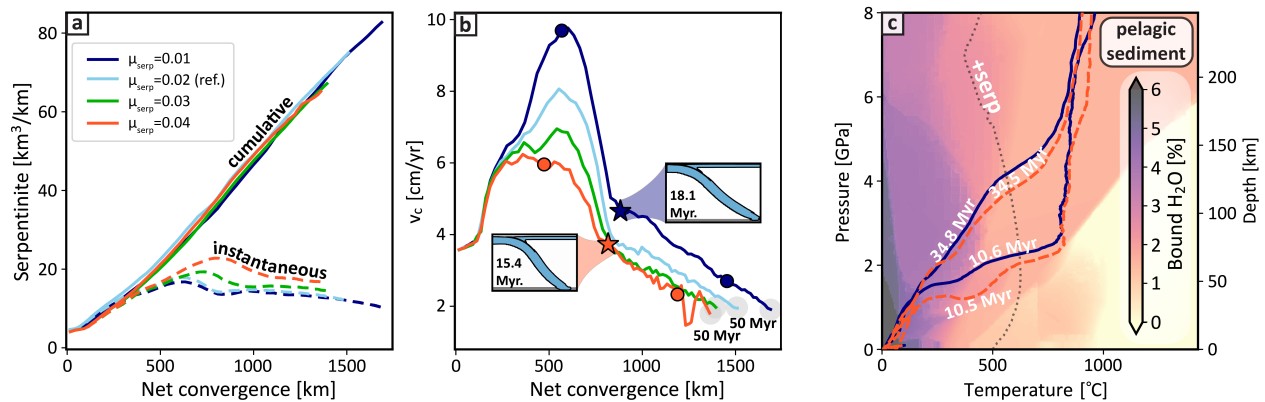


Figure 8. Kinematic and thermal evolution of models with varying the effective friction coefficient of serpentinite. (a) Ingrowth of serpentinite over the course of the model run either considering the *cumulative* serpentinite amount that was created over the duration of the entire model run or the *instantaneous* amount that indicates the amount of serpentinite that is stable at any given point in time. The difference between the two is an indication of the degree of loss of serpentinite due to coupling with the downgoing plate. (b) Changes in convergence rate for different models, showing the incorporation of serpentinite into the plate interface. Filled circles denote times when slabtop P-T conditions are plotted in c. (c) The thermal state at two snapshots in time for the strongest and weakest endmembers of serpentinite strength.

4 Discussion

4.1 Ingrowth of serpentinite

Despite exerting a critical influence on magmatism, seismicity, and the exhumation of rocks from depth, the mantle wedge remains one of the most enigmatic regions of the subduction system. In particular, the amount of water bound in hydrous minerals, particularly serpentinite, is poorly resolved, partly due to the paucity of phase changes in wedge serpentinites, which complicates efforts to precisely constrain pressure-temperature (P-T) conditions. Unlike most crustal rocks, serpentinites rarely contain minerals suitable for geochronological dating. Consequently, constraining the ingrowth of serpentinites in the mantle wedge, as well as linking them to large-scale kinematics, necessitates alternative approaches, such as geochemical analyses of arc magmas or of serpentinite mud volcanoes, geophysical techniques, or numerical modeling (e.g., Tatsumi, 1986; Tatsumi & Eggins, 1995; Fryer et al., 1999; Hyndman & Peacock, 2003). Linking mineral phase stability calculations in the downgoing plate to thermomechanical models provides a quantitative framework for evaluating serpentinite stability and the amount of water released by the slab. Our numerical models expand on this framework by integrating the thermal state of the downgoing slab, its dewatering processes, and the resulting mantle wedge hydration over time, thereby complementing petrological and geochemical techniques (Scambelluri & Tonarini, 2012; Abers et al., 2017).

This work examines the effects of the progressive ingrowth and buildup of serpentinite in the mantle wedge over the lifetime of a subduction zone (Figures 4-8). Extensive geodynamic and petrological modeling has shown that water released from the downgoing slab hydrates and serpentinitizes the forearc mantle wedge (e.g., Peacock, 1987; Wada and Wang, 2009; Abers et al., 2017; Condit et al., 2020). However, how the time-dependent ingrowth of serpentinites, and thus of a region of rheologically weak material, feed back into subduction kinematics, such as plate velocities and slab morphology, was beyond the scope of previous studies. While partial feedback has been incorporated into time-dependent numerical studies, these typically imposed kinematic boundary conditions—usually plate velocities—and hence prevent the progressive accumulation of weak serpentinite from influencing large-scale plate motions. These studies have nonetheless successfully tracked serpentinitization in the context of exhumation, wedge dynamics, magmatism, and/or decoupling of the downgoing slab from the overriding wedge (Gerya et al., 2002; Arcay et al., 2005; Nikolaeva et al., 2008; Kerswell et al., 2021).

In contrast, our dynamic models allow plate forces to evolve freely, capturing key feedback between serpentinite rheology, mantle wedge hydration, and thermal state. Previous work has already examined certain aspects of this feedback loop. For instance, rapid serpentinitization during subduction infancy has been linked to evolving temperatures in the slab (Epstein et al., 2024; Holt and Condit, 2021). The interface rheology has also been shown to modulate kinematics, including convergence rates (Behr et al., 2022). Furthermore, the weakening effect of a hydrated wedge after subduction initiation has been proposed based on force balances from kinematic models (Arcay et al., 2005). Our work validates these findings

and expands upon them with fully coupled dynamic models that allow feedback to evolve self-consistently. We demonstrate that during subduction infancy plate convergence rates are independent of serpentinite ingrowth. At later stages, however, once serpentinites have accumulated, convergence rates are influenced by the abundance of stable serpentinite (Figure 6, 8).

The extent of serpentinitization of the mantle wedge over time is a mass balance between serpentinite ingrowth and loss. Ingrowth of serpentinite depends on the amount of water released from the downgoing slab, depending on the thermal state. Early subduction zone thermal structures are predicted to broadly cool based on the exhumed rock record (Platt, 1975; Cloos, 1985; Agard et al., 2018; Penniston-Dorland et al., 2025), a trend that emerges as the downgoing slab refrigerates the forearc (Figures 4-5). This is consistent with work showing time-dependent dehydration and serpentinitization in dynamic models (Holt and Condit, 2021; Epstein et al., 2024). Serpentinitization evolves concurrently with accelerating plate convergence: plate convergence increases the most during nascent subduction and the free sinking phase. Therefore, the greatest rate of serpentinitization also occurs within the early and free sinking phases when hydrous material is subducted the most rapidly (Figure S2).

Serpentinite ingrowth is also strongly influenced by the composition of the downgoing material (Figures 4-5) that uniquely responds to the P-T conditions it encounters. Hydrated sediment produces more water that drives serpentinitization in the mantle wedge compared to metagabbros and metabasalts (Figure 5). More serpentinitization in sediment-rich models was originally shown for a sediment with an averaged sediment composition (GLOSS), which also tends to dehydrate at shallower conditions than metabasalts (Rüpke et al., 2004). Greater serpentinitization extents with more sediment are also consistent with more recent modeling (Abers et al., 2017; Epstein et al., 2024). Models without sediment experience much less serpentinitization because metabasalts contain less water (2.3 wt.% vs. 6.89 wt.%) and dehydrate outside of serpentinite stability, especially as the model matures (Figures 5-6). A global assessment of the most common subducting sediment compositions (Plank, 1998) finds pelagic sediments to be the most prevalent. Other compositions, such as carbonate-rich sediments, devolatilize and may further weaken the plate interface (Oyanagi & Okamoto, 2024), but decarbonation is beyond the scope of the current work.

Downdragging represents an additional complexity by removing serpentinite from the mantle wedge. Significant coupling between serpentinite and the downgoing slab was originally proposed as a mechanism to delay dehydration of material in or near the slab until it reaches depths necessary for generating hydrous arc magmas (Tatsumi, 1986). This is also evidenced by geochemical studies that show a striking similarity between forearc serpentinites and arc magmas (Tonarini et al., 2011; Scambelluri et al., 2019). From a mass balance perspective (Clift and Vannucchi, 2004), downdragging would prevent some forearcs from being oversaturated (Savov et al., 2007; Abers et al., 2017; Epstein et al. 2024). However, the balance between coupling and downdragging has largely been modeled in the context of subduction erosion (Angiboust et al., 2012) or the formation of a subduction channel (Schwartz et al., 2001; Gerya et al., 2002; Hilaireret et al., 2007; Krebs et al., 2007). While we only consider a limited range of serpentinite strengths in time-dependent models (Figure 7), our results agree with more extensive targeted modeling demonstrating that serpentinite downdragging depends on

serpentinite rheology (Douglas et al., 2023). We find that cases with low friction coefficients (0.01) can result in $\sim 1.5\times$ ratio of cumulative serpentinite to stable *instantaneous* serpentinite compared to the high friction case (0.04) with a ratio of $\sim 1.1\times$ at maximum convergence rates. Finally, model resolution may modestly influence the absolute quantities of downdragging. The current maximum model resolution is ~ 1.3 km, increasing model resolution to 0.7 km slightly reduces downdragging (Figure S5). Nonetheless, all cases require the formation of a rheologically weak layer of serpentinite at least several kilometers thick above the downgoing slab, which is consistent with the most recent seismological estimates using receiver functions in Cascadia (Audet et al., 2025).

Care must be taken in interpreting our models and others that make similar assumptions about the treatment of serpentinitization. The ingrowth of serpentinite is conservatively assumed to migrate directly upwards relative to the surrounding matrix, as is common in subduction modeling studies (e.g., Magni et al., 2014; Menant et al., 2019). However, channelization in the wedge or slab interface is possible as suggested by stable isotope and trace element arguments from former interface units that have since been exhumed (e.g., Penniston-Dorland and Bebout, 2016). Therefore, this assumption must be kept in mind when interpreting our absolute quantities of serpentinite. For example, after 50 Myr our models predict the ingrowth of $\sim 20\%$ percent of instantaneous serpentinite of the Epstein et al. (2024) models, which assume perfect channelization up-dip (Figure 6). Additionally, the hydration process we choose is designed to be conservative by requiring that 100% of a cell be serpentinitized before any higher cells are affected, as suggested by the presence of H_2O being the primary limiting factor to serpentinitization (Peacock, 1987). Upward-migrating swarms of earthquakes—likely indicating fluid migration—suggest that not all fluid is immediately bound upon reaching the wedge, and partial serpentinitization is possible (Davies, 1999; Halpaap et al., 2019). Incomplete reactions are also consistent with field studies of exhumed wedge serpentinites (Peacock, 1987). Finally, shear heating, not considered here, would increase the slabtop temperature (e.g., Gao & Wang, 2014; Peacock, 1992; Penniston-Dorland et al., 2015) and may increase dehydration and serpentinitization extents (Hernandez-Urbe & Palin, 2019).

4.2 Effects of serpentinite ingrowth on plate kinematics

The potential link between slab fluid release, interface strength, and hence subducting plate deformation was first recognized during the plate tectonic revolution. Studies noted that mineralogically-bound volatiles would be liberated during subduction and weaken the plate interface (Isacks and Oliver, 1968; Isacks and Molnar, 1971). A weakening of the interface affects the plate force balances and stress state along the interface. Among many factors, work has shown that the rheology and long-term coupling of the interface depends on the shear stress of the interface, with weaker material resulting in lower shear stresses over long and short timescales (Behr et al., 2022; Tassara et al., 2023).

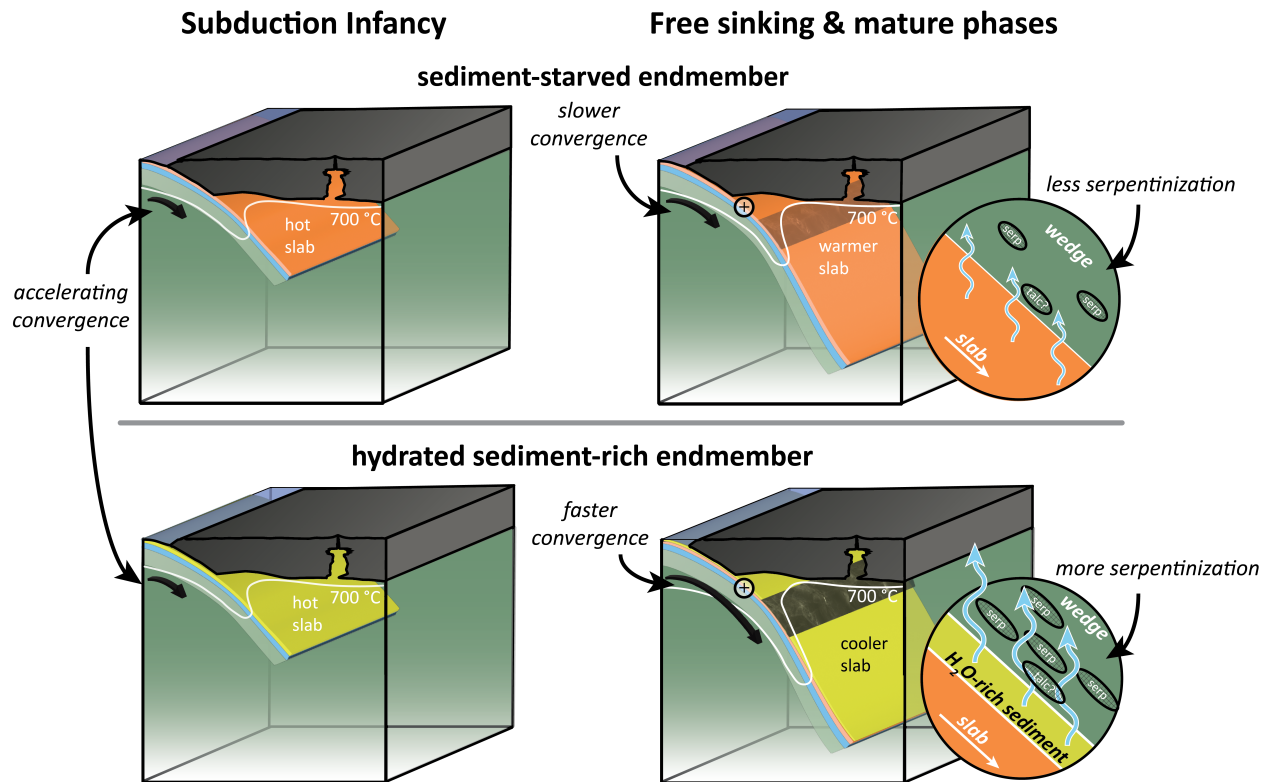


Figure 9. Summary of evolutionary feedback in our numerical results. The ingrowth of serpentinite changes the kinematics, which in turn can change the thermal state over time. All three processes are interconnected and can accentuate the influence of hydrated subducting lithologies, such as sediment, on convergence rates. Shaded regions in free sinking/mature phase correspond to regions of active serpentinization.

An array of processes, may influence plate kinematics, including the lubrication the plate interface. In particular, the interface may be lubricated by sediment (Lamb, 2006; Jagoutz et al., 2015; Behr and Becker, 2018) or ingrowing serpentinite (Wada & Wang, 2008). In our models, the accumulation of serpentinite along the plate interface changes plate kinematics, especially convergence rates, as the model matures (Figure 9). During subduction infancy, convergence proceeds more slowly (<5 cm/yr) because serpentinite has not yet accumulated to sufficient degrees to change the interface stress state. During the following free-sinking phase, the accumulated weak serpentinite causes reduced shear stresses along the interface and promotes faster convergence rates (Figure 5). More hydrated lithologies, such as sediment, can promote decoupling by forming more serpentinite as they dehydrate; a weaker interface need not arise from the effects of sediment lubrication alone (Lamb, 2006, Jagoutz et al., 2015; Behr and Becker 2018). Therefore, sediment may exert an even more pronounced influence on plate speeds than previously proposed.

A weaker interface has also been linked to the formation of a subduction channel, which may exhume rocks from depth (Shreve & Cloos, 1986). Buoyant serpentinite has invoked to exhume negatively buoyant lithologies, such as eclogites, from mantle depths (Schmitz et al., 2001, Gerya et al., 2002). Because we find time-dependent ingrowth of serpentinite, we expect that the ability for a serpentinite or serpentinite-mélange to entrain and exhume other

lithologies may also be time-dependent: more mature subduction zones may have more serpentinite available to exhume lithologies from depth. As the subduction matures, the progressive ingrowth of serpentinite weakens the interface more, meaning that serpentinite might be a more viable mechanism of exhumation after subduction reaches maturity. Nonetheless, we do not observe large scale exhumation in our models: additional mechanisms such as more extensive surface erosion or the termination of subduction may promote such exhumation (c.f., England and Smye, 2023).

Serpentinite is expected to be common along the plate interface, and therefore we expect that interface weakening may be a general phenomenon, rather than being tied to a specific region. A specific comparison to plate speeds may be challenging because of other factors that affect subduction that include plate age (Isacks, 1968; Stegman, 2015), plate and trench geometry (Jagoutz et al., 2015), overriding plate deformation (Lallemand, 2005), and forces associated with mantle flow (Ueda et al., 2008) and interactions with the 660-km plate discontinuity. Further work could expand this study by considering metasomatism. Weak hydrous minerals that accumulate via metasomatism (Behnsen and Faulkner, 2012; Lindquist et al., 2023), such as talc (Peacock and Wang, 2021), likely also contribute to low shear stresses along the plate interface.

5 Conclusions

This study tracked the ingrowth of serpentinite in a subduction system in dynamical modals to determine how the progressive serpentinitization of the mantle wedge would cause feedback between kinematics, dehydration, and thermal state over time. We found that hydration state initially exerts a minor influence on convergence rates. However, after subduction infancy slabs with hydrated sediments produce more serpentinite, leading to increased convergence rates ($\sim 30\%$) compared to cases with sediment that did not have any bound H_2O . The magnitude of weakening due to serpentinite strongly depends on the friction parameters of serpentinite. A case where the strength of serpentinite is 4x weaker than the downgoing crust exhibits increased convergence rates by 2 cm/yr during the free sinking phase and 1 cm/yr during the mature phase compared to when the crust and serpentinite have identical strength properties. Our work also offers a way to link two commonly proposed hypotheses in subduction zones, weakening of the interface due to the subduction of sediments and weakening of the interface due to the ingrowth of hydrous minerals. Altogether, the lubricating effect of serpentinite may act in concert with the lubricating effect of sediment.

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Open Research

AGU requires an Availability Statement for the underlying data and code needed to understand, evaluate, and build upon the reported research at the time of peer review and

publication. Additionally, authors should include an Availability Statement for the software and other research products that have a significant impact on the research. Details and templates are in the [Availability Statement](#) section of the Data & Software for Authors Guidance. For physical samples, use the IGSN persistent identifier, see the [International Geo Sample Numbers](#) section.

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